Machine Learning – Hypothesis Space

Given information

• There are p attributes: A_1, A_2, ..., A_p.

• Each attribute A_i has m_i possible values.

• The classification concepts are C = {0, 1}.

• A hypothesis is a conjunction of restrictions on attributes.

• Each attribute can have one of its m_i specific values or a wildcard (?) meaning any value.

• The hypothesis space also contains the empty hypothesis ⊥, which classifies every example as 0.

(a) Determine the number n(p) of all possible examples

An example consists of one value for each of the p attributes. Using the multiplication principle, attribute A_1 has m_1 choices, attribute A_2 has m_2 choices, and so on, until attribute A_p has m_p choices.

$$\mathbf{n(p) = \prod_{i=1}^{p} m_i}$$

Answer: The total number of possible examples is the product of the number of values of all attributes.

(b) Determine the number $|H_p|$ of different hypotheses

For every attribute A_i, there are m_i possible specific values and one wildcard (?) option. Therefore, the number of choices for each attribute is $m_i + 1$.

Since a hypothesis is a conjunction of restrictions, the total number of hypotheses is:

$$\mathbf{|H_p| = 1 + \prod_{i=1}^{p} (m_i + 1)}$$

The additional 1 represents the empty hypothesis ⊥.

(c) Add an additional attribute A_{p+1} with m_{p+1} values

1. Number of possible examples

Previously: $n(p) = \prod_{i=1}^{p} m_i$
After adding the new attribute: $n(p+1) = \prod_{i=1}^{p+1} m_i$

$$\mathbf{\text{Recursion formula: } n(p+1) = n(p) \cdot m_{p+1}}$$

2. Number of different hypotheses

Previously: $|H_p| = 1 + \prod_{i=1}^{p} (m_i + 1)$
After adding the new attribute: $|H_{p+1}| = 1 + \prod_{i=1}^{p+1} (m_i + 1)$

$$\mathbf{\text{Recursion formula: } |H_{p+1}| = 1 + (|H_p| - 1)(m_{p+1} + 1)}$$

Final answers

Question	Answer
(a) Number of examples	$n(p) = \prod_{i=1}^{p} m_i$
(b) Number of hypotheses	$\|H_p\| = 1 + \prod_{i=1}^{p} (m_i + 1)$
(c) Recursion for examples	$n(p+1) = n(p) \cdot m_{p+1}$
(c) Recursion for hypotheses	$\|H_{p+1}\| = 1 + (\|H_p\| - 1)(m_{p+1} + 1)$